

Kinematics & Control

Differential Drive Line-Follower Robot

1. System Parameters

The kinematic model is defined by the physical geometry of the robot and the motion of its actuators.

Symbol	Description
r	Wheel radius
L (or $2b$)	Wheelbase / Track width (center-to-center distance)
ω_R, ω_L	Angular velocities of the right and left wheels (rad/s)
v	Linear velocity of the robot center (m/s)
ω	Angular velocity of the robot body (rad/s)
θ	Orientation (heading) in the global frame

2. Forward Kinematics (Local Frame)

Forward kinematics maps the wheel velocities to the robot's body velocities (v, ω). Using the wheelbase $L = 2b$:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} r/2 & r/2 \\ r/L & -r/L \end{bmatrix} \begin{bmatrix} \omega_R \\ \omega_L \end{bmatrix}$$

3. Unicycle Model (Global Frame)

To determine the robot's position in the global coordinate system (x, y, θ) , the body velocities are rotated into the global frame:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & 0 \\ \sin(\theta) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

4. Odometry Integration

By combining the above matrices and integrating over a time step Δt , we estimate the robot's pose (x, y, θ) over time:

$$P(t) = P(t - \Delta t) + \Delta t \cdot J(\theta) \cdot \Phi^*$$

$$\begin{bmatrix} x(t) \\ y(t) \\ \theta(t) \end{bmatrix} = \begin{bmatrix} x(t-\Delta t) \\ y(t-\Delta t) \\ \theta(t-\Delta t) \end{bmatrix} + \Delta t \begin{bmatrix} (r/2)\cos(\theta) & (r/2)\cos(\theta) \\ (r/2)\sin(\theta) & (r/2)\sin(\theta) \\ r/L & -r/L \end{bmatrix} \begin{bmatrix} \omega_R \\ \omega_L \end{bmatrix}$$

5. Inverse Kinematics

To achieve a target linear velocity and angular velocity, calculate the required wheel speeds:

$$\begin{bmatrix} \omega_R \\ \omega_L \end{bmatrix} = (1/r) \begin{bmatrix} 1 & L/2 \\ 1 & -L/2 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

6. Controller Design (PID)

For line following, a controller adjusts the turning rate (ω) to minimize the lateral error (e) from the line.

$$\omega(t) = K_p e(t) + K_i \int e(t)dt + K_d (de/dt)$$

- **Proportional (P):** Immediate correction based on current error.
- **Integral (I):** Corrects steady-state errors (e.g., motor bias).
- **Derivative (D):** Dampens oscillations by predicting future error.

Implementation Note: On microcontrollers like the STM32, ensure the integration step Δt is constant using a timer interrupt to avoid odometry drift and unstable PID behavior.